

Mathematical Derivation of the Fisher Information Matrix

1. Problem setup

Have a pixelated detector image represented by a random vector $Y = (Y_1, Y_2, \dots, Y_K)^T$, where K is the number of pixels. The parameter vector is

$$\theta = (x_0, y_0, z_0)^T \in \mathbb{R}^3 \quad (1)$$

where (x_0, y_0, z_0) is the true 3D position of a single fluorescent emitter.

The mean detected signal in pixel k is modeled as

$$\mu_k(\theta) = N \cdot q_k(\theta) \quad (2)$$

where N is the total expected photon count from the emitter, and $q_k(\theta)$ is the pixel-integrated, normalized point-spread function (PSF):

$$q_k(\theta) \geq 0, \quad \sum_{k=1}^K q_k(\theta) = 1 \quad (3)$$

In continuous form, q_k is the fraction of emitted photons collected in pixel k :

$$q_k(\theta) = \iint_{\text{pixel } k} p(u, v \mid x_0, y_0, z_0) du dv \quad (4)$$

where p is the (unnormalized) PSF density on the detector plane.

2. Poisson observation model

Each pixel count is an independent Poisson random variable:

$$Y_k \sim \text{Poisson}(\lambda_k(\theta)), \quad k = 1, \dots, K \quad (5)$$

with mean

$$\lambda_k(\theta) = \mu_k(\theta) + b_k \quad (6)$$

where $b_k \geq 0$ is the effective background in pixel k . In the implementation, $b_k = b$ is constant across pixels.

3. Log-likelihood function

The joint probability mass function of Y given θ is

$$P(Y \mid \theta) = \prod_{k=1}^K e^{-\lambda_k(\theta)} \lambda_k(\theta)^{Y_k} / Y_k! \quad (7)$$

The log-likelihood is therefore

$$\ell(\theta; Y) = \sum_{k=1}^K [Y_k \log \lambda_k(\theta) - \lambda_k(\theta) - \log(Y_k!)] \quad (8)$$

4. Score function

The score (gradient of the log-likelihood) with respect to θ_i is

$$\partial \ell / \partial \theta_i = \sum_{k=1}^K (Y_k / \lambda_k(\theta) - 1) \cdot \partial \lambda_k(\theta) / \partial \theta_i \quad (9)$$

Since $\lambda_k = \mu_k + b_k$ and b_k does not depend on θ ,

$$\partial \lambda_k / \partial \theta_i = \partial \mu_k / \partial \theta_i \quad (10)$$

Using $\mu_k = N q_k(\theta)$ from Eq. (2),

$$\partial \mu_k / \partial \theta_i = N \cdot \partial q_k(\theta) / \partial \theta_i \quad (11)$$

5. Fisher information matrix — definition

The Fisher information matrix is defined as

$$I(\theta) = E_Y [(\partial \ell / \partial \theta) (\partial \ell / \partial \theta)^T] \quad (12)$$

Equivalently, under regularity conditions,

$$I_{\{ij\}}(\theta) = - E_Y [\partial^2 \ell / (\partial \theta_i \partial \theta_j)] \quad (13)$$

6. Fisher information for Poisson pixels — step-by-step derivation

Step 6.1 Second derivative of the log-likelihood

Differentiating Eq. (9) with respect to θ_j :

$$\partial^2 \ell / (\partial \theta_i \partial \theta_j) = \sum_k [-(Y_k / \lambda_k^2) \cdot (\partial \lambda_k / \partial \theta_i) (\partial \lambda_k / \partial \theta_j) + (Y_k / \lambda_k - 1) \cdot \partial^2 \lambda_k / (\partial \theta_i \partial \theta_j)] \quad (14)$$

Step 6.2 Take expectation

For Poisson Y_k , $E[Y_k] = \lambda_k$. Hence $E[Y_k / \lambda_k] = 1$ and

$$E[\partial^2 \ell / (\partial \theta_i \partial \theta_j)] = \sum_k [-(1 / \lambda_k) (\partial \lambda_k / \partial \theta_i) (\partial \lambda_k / \partial \theta_j) + (\partial^2 \lambda_k / (\partial \theta_i \partial \theta_j)) \cdot E[Y_k / \lambda_k - 1]] \quad (15)$$

The second term vanishes because $E[Y_k / \lambda_k - 1] = 0$. Therefore,

$$I_{\{ij\}}(\theta) = \sum_{k=1}^K (1 / \lambda_k(\theta)) \cdot (\partial \lambda_k(\theta) / \partial \theta_i) (\partial \lambda_k(\theta) / \partial \theta_j) \quad (16)$$

Step 6.3 Substitute the localization model

With $\lambda_k = N q_k(\theta) + b$ and $\mu_k = N q_k(\theta)$, we have $\partial \lambda_k / \partial \theta_i = N \partial q_k / \partial \theta_i$. Eq. (16) becomes

$$I_{\{ij\}}(\theta) = N^2 \cdot \sum_{k=1}^K [1 / (N q_k(\theta) + b)] \cdot (\partial q_k / \partial \theta_i) (\partial q_k / \partial \theta_j) \quad (17)$$

In matrix form, with gradient vector $\mathbf{g}_k(\theta) = \nabla_{\theta} q_k(\theta) \in \mathbb{R}^3$:

$$I(\theta) = N^2 \cdot \sum_{k=1}^K [\mathbf{g}_k(\theta) \mathbf{g}_k(\theta)^T] / (N q_k(\theta) + b) \quad (18)$$

This is the formula implemented numerically in the project.

7. Cramér–Rao lower bound

The Cramér–Rao inequality states that for any unbiased estimator $\hat{\theta}$ of θ ,

$$\text{Cov}(\hat{\theta}) \geq I(\theta)^{-1} \quad (19)$$

For the scalar coordinates x_0, y_0, z_0 :

$$\text{Var}(\hat{x}_0) \geq [I(\theta)^{-1}]_{11} \quad (20)$$

$$\text{Var}(\hat{y}_0) \geq [I(\theta)^{-1}]_{22} \quad (21)$$

$$\text{Var}(\hat{z}_0) \geq [I(\theta)^{-1}]_{33} \quad (22)$$

The Cramér–Rao lower bounds on localization precision are

$$\sigma_{x}^{\text{CRLB}} = \sqrt{[I(\theta)^{-1}]_{11}} \quad (23)$$

$$\sigma_{y}^{\text{CRLB}} = \sqrt{[I(\theta)^{-1}]_{22}} \quad (24)$$

$$\sigma_{z}^{\text{CRLB}} = \sqrt{[I(\theta)^{-1}]_{33}} \quad (25)$$

8. PSF models substituted into $q_k(\theta)$

8.1 Astigmatic PSF

The continuous PSF is a separable elliptical Gaussian with z-dependent widths:

$$p(u, v | z_0) \propto \exp[-(u-x_0)^2 / (2 \sigma_x(z_0)^2) - (v-y_0)^2 / (2 \sigma_y(z_0)^2)] \quad (26)$$

$$\sigma_x(z_0) = \sigma_{x0} \sqrt{1 + ((z_0+c)/d)^2 + a_x((z_0+c)/d)^3 + b_x((z_0+c)/d)^4} \quad (27)$$

$$\sigma_y(z_0) = \sigma_{y0} \sqrt{1 + ((z_0-c)/d)^2 + a_y((z_0-c)/d)^3 + b_y((z_0-c)/d)^4} \quad (28)$$

Pixel values q_k are obtained from Eq. (4) by integrating over each pixel.

8.2 Double-helix PSF (parametric model)

The DH-PSF is modeled as two equal Gaussian lobes rotating with axial position:

$$\varphi(z_0) = \varphi_{\text{max}} \cdot (z_0 / z_{\text{ref}}) \quad (29)$$

$$p(u, v | \theta) = (1/2) [G(u-x_1, v-y_1; \sigma_{\ell}) + G(u-x_2, v-y_2; \sigma_{\ell})] \quad (30)$$

$$(x_1, y_1) = (x_0 + (s/2)\cos \varphi, y_0 + (s/2)\sin \varphi) \quad (31)$$

$$(x_2, y_2) = (x_0 - (s/2)\cos \varphi, y_0 - (s/2)\sin \varphi) \quad (32)$$

where G is a normalized 2D Gaussian of width σ_ℓ and s is the lobe separation. This is a parametric approximation (Badieirostami et al., 2010).

8.3 Derivatives $\partial q_k / \partial \theta_i$

Because $q_k(\theta)$ is obtained by pixel integration (and lobe rotation for DH-PSF), closed-form derivatives are not used. Numerically,

$$\partial q_k / \partial x_0 \approx [q_k(x_0 + \Delta x, y_0, z_0) - q_k(x_0 - \Delta x, y_0, z_0)] / (2\Delta x) \quad (33)$$

and analogously for y_0 and z_0 . These are substituted into Eq. (17).